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Embedding machine learning techniques into a conceptual model to improve monthly runoff simulation: A nested hybrid rainfall-runoff modeling

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ABSTRACT

One of the frequently adopted hybridizations within the scope of rainfall-runoff modeling rests on directing various outputs simulated from the conceptual rainfall-runoff (CRR) models to machine learning (ML) techniques. In those coupled model exercises, after the parameter calibrations of the CRR models are made, their specific outputs constitute auxiliary inputs for the ML model training. However, in this parallel hybridization comprising two consecutive processes, performing the cascade calibration of CRR and ML models increases the computational complexity. Moreover, the mutual interaction between the parameters governing CRR and ML models is also not considered. In this study, to cope with the handicaps mentioned, artificial neural networks (ANN) and support vector regression (SVR) were separately embedded into a monthly lumped CRR model. The dynamic water balance model (*dynwbm*) was preferred as the CRR model. Then, all free parameters within these nested hybrid models were calibrated simultaneously. The ML parts within the nested schemes manipulate various output variants derived with three conceptual parameters for monthly runoff simulation. These new hybrid models equipped with an automatic calibration algorithm were applied at several locations in the Gediz River Basin of western Turkey. The performance measures regarding mean and high flows indicated that the nested hybrid models outperformed the standalone models (i.e., *dynwbm*, ANN, and SVR) and coupled model variants. Thus, the credibility of a novel modeling strategy, which takes advantage of the supplementary strengths of a conceptual model and different ML techniques, was demonstrated.

1. Introduction

1.1. General review

Revealing the rainfall-runoff relationship controlled by the rainfall regime and geomorphic properties of basins has been the main topic among hydrological studies for a long time. The mechanism of transformation of rainfall to runoff is a rather complicated and nonlinear process due to the hydrological components that are distributed heterogeneously in the basin and the spatial-temporal variations of these components (Song et al., 2012). Rainfall-runoff models developed within this context are employed operationally to provide reliable streamflow forecasts to flood control, reservoir operation studies, and optimal water allocation. The development of these models also

improves our understanding of the hydrological processes continually occurring in the watersheds (Clark et al., 2011).

Regardless of the structural variety of these models, they can be typically divided into three fundamental types, namely, physically-based models, parametric conceptual models, and black box models or system theoretical models (Dooge 1977; Chen and Adams, 2006a; 2006b; Song et al., 2012). These models employ spatially distributed input information such as soil characteristics, topography, and vegetation. Besides, intensive computational effort is required to parameterize and validate physically-based models running with high quality data (Seyfried and Wilcox, 1995). In the flood forecasting scope, operational studies are frequently performed with physically-based hydrological models.

On the other hand, when good simulation accuracy is aimed with less

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data or parameters, conceptual rainfall-runoff (CRR) models and black box models are practical options (Nourani and Mano, 2007; Humphrey et al., 2016). Among them, CRR models, whereby the entire basin is usually considered homogenous, have been widely used in research studies. They can represent interconnecting physical elements and various hydrological processes through a series of conceptual soil moisture and groundwater stores with simplified mathematical expressions (Chen and Adams, 2006b; Noori and Kalin, 2016). It is assumed in CRR models that the rainfall is uniformly distributed over the basin. Besides, their parameters are defined within the lumped manner despite the variations in the spatial distribution of parameters (Song et al., 2012). Moreover, the effective implementation of CRR models depends on their calibration sensitivity. In this regard, the qualification of the automatic optimization algorithm to be evaluated in the calibration stage is also essential for accurate modeling (Arsenault et al., 2014).

The other kind is black box modeling that simulates runoff series from hydrometeorological variables used as inputs. It covers several statistical techniques ranging from conventional regression methods to versatile machine learning (ML) algorithms (i.e., data-driven models). It has been stated in the literature that these models, which do not need detailed knowledge about the hydrological process, can be good alternatives to CRR models (Nourani and Mano, 2007; Ren et al., 2018). Undoubtedly, among the other statistical data-driven methods, artificial neural networks (ANNs) models have been used as a nonlinear regression approach since the mid-90 s. In the practice, having an ability to mimic the hydrological process utilizing various observations alone, offering relatively more flexible modeling with nonlinear mapping, and compensating for the missing hydrometeorological information are the main advantageous aspects of ANNs (Zhang and Govindaraju, 2003).

In the domain of rainfall-runoff modeling, various studies have performed direct comparisons of ANNs and CRR models. Some studies emphasized that ANNs performed better than conceptual models (Dibike and Solomatine, 2001; Jeong and Kim, 2005). On the other hand, other studies underlined that CRR modeling would be more robust in scenario-based forecasting and not as prone to overfitting as ANNs (Gaume and Gosset, 2003; Daliakopoulos and Tsanis, 2016). Wang et al. (2009) and Maity et al. (2010) emphasized that the support vector regression (SVR) is a powerful alternative that can overcome the weaknesses of ANNs (i.e., the difficulty in determining the number of nodes in the hidden layer and the possibility of over-training due to the numerous number of weights). In this content, other data-driven techniques (e.g., multivariate adaptive regression splines, genetic programming, and the adaptive neural-based fuzzy inference system) have also attracted the interest of hydrologists. However, ANN and SVR can be preferred over them because they have provided relatively robust results in many water resources exercises (Tongal and Booij 2018). Therefore, it was decided to employ both techniques for capturing nonlinear hydrological relationships in this study.

1.2. Coupled conceptual-machine learning modeling framework

In fact, in most cases, data-driven models such as ANN and SVR cannot compete with CRR models without the use of antecedent rainfall and antecedent runoff series. Adding antecedent observations to the input set is a common practice in order that conveying the impacts of soil moisture-groundwater storage (or initial basin wetness) into the modeling can be better represented (Ancil et al., 2004; Okkan and Serbes, 2012; Robertson et al., 2013; Humphrey et al., 2016; Kumanlioglu and Fistikoglu, 2019). However, the models trained with such antecedent runoff series constitute an impediment for streamflow projection studies (e.g., exercises related to downscaling of global climate model outputs to runoff) under climate change scenarios because such an input series cannot be provided for future climatic conditions (Okkan and Inan, 2015; Kumanlioglu and Fistikoglu, 2019). Additionally, Robertson et al. (2013) have highlighted that using antecedent streamflow and rainfall observations is limited to describing catchment

dynamics for two reasons.

(i) When soil moisture and groundwater storages reach their full capacities, runoff can proceed to increase to levels beyond those that reflect basin moisture storage. Therefore, in case of excessive antecedent runoff observations, subsequent outputs may be higher than the actual storage levels would cause.

(ii) When a basin is getting wet, antecedent runoff does not immediately react to antecedent rainfall, much rather soil moisture and groundwater storages refill primarily. Under this situation, antecedent runoff values may underestimate the actual soil moisture conditions and lead to getting relatively low runoff estimations.

Although there are new attempts made up of several statistical exercises where ML models are hybridized with nonlinear time series models such as fractionally autoregressive integrated moving average, self-exciting threshold autoregressive, and generalized autoregressive conditional heteroscedasticity to improve streamflow simulation (e.g., Mehdizadeh et al., 2019; Fathian et al., 2019), neither standalone ML techniques nor entirely statistical hybrid models have been preferred in operational studies (see Abrahart et al., 2012). Because the related output is characterized by the extraction of information implicitly included in a set of measured data, without considering the physical processes occurring within the system. Therefore, some researchers have sought other ways to combine the strengths of the CRR models and ML techniques as they recognize the ability of traditional conceptual models to represent the physical properties of the basins and the dominance of the data-driven models in exploring the nonlinear relationship between input and output series. In the literature, several approaches are existed to address the integration of CRR models and ML techniques. Apart from some exceptions, encountered applications can be evaluated in three groups. According to Ancil et al. (2004) and Noori and Kalin (2016), those studies can also be referred to as *parallel hybridization* or *coupled models*.

The models that can be evaluated in the first group contain the assessment of ML models for the statistical post-processing of CRR model outputs. Some studies within this group refer to multi-model ensemble strategies (e.g., Shamseldin et al., 1997; Ancil et al., 2003; Kim et al., 2006). Rather than concentrating on outputs derived from a single rainfall-runoff model, these studies consider nonlinearly weighted outputs of several rainfall-runoff models in the expectation of getting better runoff simulation and minimizing associated uncertainties of individual models (Humphrey et al., 2016). Besides, modeling practices that update the simulated runoff series obtained from a single CRR model and therefore perform error-correction (EC) can also be included in this group (e.g., Shamseldin and O'Connor, 2001; Ghaith et al., 2020).

The models in the second group simulate the total runoff at the entire basin outlet by the ML-based nonlinear routing performed to runoff components generated from individual sub-basins. In those studies, following deriving modeled runoff series from the sub-catchments using lumped or semi-distributed CRR models, the contribution of those from each sub-catchment to the total runoff is determined by ML-based models instead of a classical linear superposition. For example, the ML-based runoff routing process was successfully coupled with the simulations of the TANK model (Chen and Adams, 2006a), variable infiltration capacity model (Meng et al., 2016), the XAJ model (Song et al., 2012), and the soil moisture accounting and routing model (Chen and Adams, 2006b).

On the other side, the models in the third group, which are relatively sophisticated, manipulate various outputs from CRR models (simulated data associated with hydrological process components) to reinforce the predictor set for ML models (Humphrey et al., 2016). One of the most remarkable studies on this subject was performed by Ancil et al. (2004). They incorporated the simulated soil moisture index values obtained from the GR4J model with predefined parameters into the ANNs with a view to improving short-term streamflow simulation. While attempts to utilize simulated soil moisture provided promising results, they concluded that an antecedent precipitation index used as an additional

input did not significantly contribute to ANN performance. Inspired by their work, other researchers have recently introduced similar implementations. For example, [Humphrey et al. \(2016\)](#) used simulated soil moisture data from the GR4J model to represent initial basin conditions in the Bayesian neural network model exerted to monthly streamflow simulation for some South Australia basins. [Kumanlioglu and Fistikoglu \(2019\)](#) utilized the production storage outputs (e.g., actual evapotranspiration, percolation, and effective rainfall) of the same conceptual model to strengthen the performance of ANNs in daily runoff simulation for the Gediz River Basin located in western Turkey. The coupling same conceptual model with the Elman recurrent neural network was also carried out by [Tian et al. \(2018\)](#).

Moreover, [Nilsson et al. \(2006\)](#) and [Ren et al. \(2018\)](#) employed the HBV model outputs, including soil moisture, snowmelt, and glacier-melt runoff, as auxiliary ANN inputs to better simulate monthly runoff. Also, [Noori and Kalin \(2016\)](#) and [Essenfelder and Giupponi \(2020\)](#) incorporated simulated data obtained from the Soil and Water Assessment Tool (SWAT) into ANN to enhance streamflow simulation. SWAT-ANN coupled approach was evaluated for water quality prediction as well ([Noori et al., 2020](#)). Furthermore, there are also studies in which the streamflow simulation success of ML models is increased by separating components such as baseflow and stormflow through the process-oriented methods such as the curve number approach of the Soil Conservation Service ([Isik et al., 2013](#)) and the single-pass digital filter method ([Tongal and Booi, 2018](#)).

1.3. An alternative approach: nested hybridization

In all those studies attributed in the third group, CRR models are primarily calibrated, and then specific outputs of them are handled as new inputs within ML models. That is, both calibration processes are operated alternately, and the ML-based part is inactive before the CRR model calibration is completed. [Antil et al. \(2004\)](#), who suggested parallel hybridization in their study, emphasized that serial (we called it nested) hybrid modeling, where both approaches are intertwined in one scheme, would be more hydrologically expressive. Accordingly, if nested hybridization, where all defined parameters are optimized together within the same iteration steps, is not preferred, the mutual interactions between CRR model parameters and ML parameters will become inherently restricted. Additionally, the computational intensity will increase with two consecutive modeling processes as a matter of course. In the study presented, to test the credit of nested hybridization mentioned, the embedding of two ML techniques into a monthly CRR model termed as *dynamic water balance model (dynwbm)* has been provided. Then, all free parameters (*dynwbm* parameters and related ML parameters) have been simultaneously calibrated in this intertwined structure. Of course, while doing this, it is crucial to check whether the model pairs to be hybridized are in harmony with each other and substantially eliminate their weaknesses (i.e., over-parameterization and sensitivity to initial conditions). In this concept, hybridizations have been first performed with familiar multilayer ANN. However, in this case, the degree of freedom of the nested hybrid model will be increased, and ANN parameters will become much more influence on the total variance uncertainty. We know that integrating ANNs into a lumped conceptual model at this point contradicts the judgment that the parsimonious model is relatively practical. Hence, as an alternative to the ANN, we have thought that over-parametrization can be avoided using least-squares support vector regression (SVR), which has only two parameters to be tuned within the hybrid scheme. To the best of our knowledge, these kinds of nested hybrid approaches made up of ML algorithms have not yet been proposed for the scope of the monthly rainfall-runoff modeling. As is known, according to a comprehensive review conducted by [Abraham et al. \(2012\)](#), the interest for ML-based models has not yet percolated to operational agencies (e.g., governments, private companies). In this regard, as a robust alternative to the sole ML models, the nested hybrid modeling strategy proposed may be a coherent way to

link to operational studies in terms of their potential to better capture the basin dynamics.

In this study, to diagnose the benefits of the nested hybridization implementations, a comparison has been undertaken on relative performances obtained from all candidate models for producing monthly runoff simulations at streamflow gauging stations located in Gediz Basin in western Turkey. In the comparison step, runoff simulation performances of the sole models (i.e., sole ANN, sole SVR, and *dynwbm*) and parallel hybrid models (i.e., simple error-correction (EC) variants and coupled model (CM) variants) have been analyzed against nested hybrid models. Such research disclosing the pros and cons of different hybridization schemes against each other has not been addressed so far. The remainder of the study is arranged as follows: first, the description of the data and methodology is presented; the results derived from the adopted models are then discussed; and ultimately, conclusions are drawn.

2. Study area and data

In the study, the Gediz Basin, which highly ensures the agricultural water demands of the Aegean region of Turkey, was chosen as the study area because there is still growing demand for water use over the basin. Due to the water use competitions happened in the recent past, some problems involving water shortages, pollution, salinity, and conflicts between agricultural and industrial users have occurred. Furthermore, urban areas in the basin are also projected to expand by 11% by 2030 ([De Voogt et al., 2000](#)). Accordingly, it is inevitable to experience similar matters in the near future. Several studies have been carried out and are still being implemented in this region where the typical Mediterranean climate is much more prominent ([Choudhury, 2000](#); [De Voogt et al., 2000](#); [Kite, 2000](#); [Karatas et al., 2009](#)).

Since the agricultural water demands are determined by monthly hydrometeorological data in institutional operations in Turkey ([Karatas et al., 2009](#)), the accurate simulation of monthly time-scaled streamflow series is essential for planning the allocation of water to the irrigation areas and questioning the adequacy of water supply in the agricultural-operated dam reservoirs. In this respect, it was decided to analyze models operating with monthly time-scaled data. The meteorological stations over the basin that measure monthly precipitation, temperature, and relative humidity data are operated by the Turkish State Meteorological Service. Since wind speed and solar radiation data, which are not measured regularly, constitute the basis for the empirical Penman-Monteith equation-based PET calculation, data were compensated by using ERA-Interim reanalysis data served by the European Center for Medium-Range Weather Forecasts. Besides, nine streamflow gauging stations operated by the General Directorate of State Hydraulic Works are located at specific key points of the basin. [Table 1](#) provides general information about streamflow gauging stations considered. Derekoy, Topuzdamlari, Borlu, and Acisu stations are on the tributaries feeding the Demirkopru Dam reservoir in the northeast of the basin. This reservoir and Marmara Lake, located in the center of the basin, widely supply the water demands for the dominant crops like grapes and cotton and other grown crops including maize, various vegetables, and fruits. The Marmara Lake, which is not on any stream, is fed by the water

Table 1
General information about streamflow gauging stations.

Flow Gauging Station	Station Code	Stream/Branch	Drainage area (km ²)
Muradiye	D05A25	Gediz	15,849.0
Hacihidir	D05A28	Gordes	808.2
Taytan	D05A31	Alasehir	2,513.0
Hacihaliller	D05A38	Nif	854.0
Kayalioglu	E05A09	Medar	901.6
Derekoy	E05A14	Selendi	689.6
Topuzdamlari	E05A15	Deliinis	739.6
Borlu	E05A22	Demirci	818.8
Acisu	E05A23	Gediz	3,272.4

released from the Demirkopru reservoir. The water amount over the Adala channel capacity and Alasehir streamflows are also conveyed to the lake. The Gordes reservoir fed by Gordes streamflows is also essential in providing freshwater to Izmir province. The locations of the streamflow and meteorological stations are denoted in Fig. 1. Except for the Muradiye Bridge station, which is the downstream station having the largest drainage area, various sub-basins deployed on the main tributary are specified in this figure. Those subbasins denoted are called by the name of the stations. For the drainage area of each streamflow gauging station, monthly runoff in millimeters, Thiessen-weighted precipitation data (P), and the areal mean PET data were compiled for the 1981–2010 water year period that implies current reference climate period. In this period, the mean annual P and PET regime for the whole basin are around 525 mm and 1200 mm, respectively. Besides, *semi-arid* and *dry sub-humid* climatic zones were found to be dominant for representative drainage areas of streamflow gauging stations. As shown in Fig. 2, even though the precipitation regime observed in the subbasins does not differ much, the variabilities concerning mean annual temperature, PET, and runoff are more distinctive.

3. Methodology

As mentioned in the first section, it is expected that the nested hybrid models would take advantage of the strengths of the familiar rainfall-runoff modeling approaches to get more reliable runoff simulations. Prior to elaborating on the nested hybrid models, this section briefly introduces *dynwbm* and two ML techniques to be evaluated in the hybridization phase. Thereafter, the details about how the nested hybrid models are constituted are presented. Besides, parallel hybrid models (CM and EC models) compared with nested ones are then schematized. Finally, the optimization algorithm preferred in the calibration of all models evaluated in the study and the criteria used in the performance assessment is addressed under this section.

3.1. Dynwbm

The simulated runoff outputs derived from complex CRR models with many parameters can widely match observations during the cali-

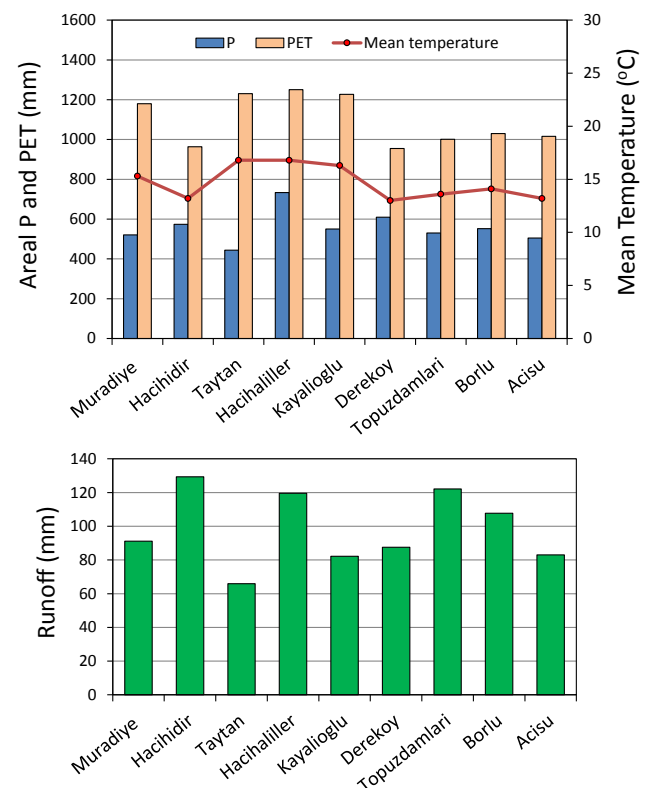


Fig. 2. Mean annual precipitation, temperature, PET, and runoff values observed in representative drainage areas of streamflow gauging stations.

bration period, but these kinds of models can be prone to suffer from over-parametrization through the validation period considered (Beven 1989). Therefore, the models attributed parametrically as parsimonious have been found much more practical (Perrin et al., 2003). Despite the existence of many CRR models, the dynamic water balance model (*dynwbm*) proposed by Zhang et al. (2008) was selected in this study

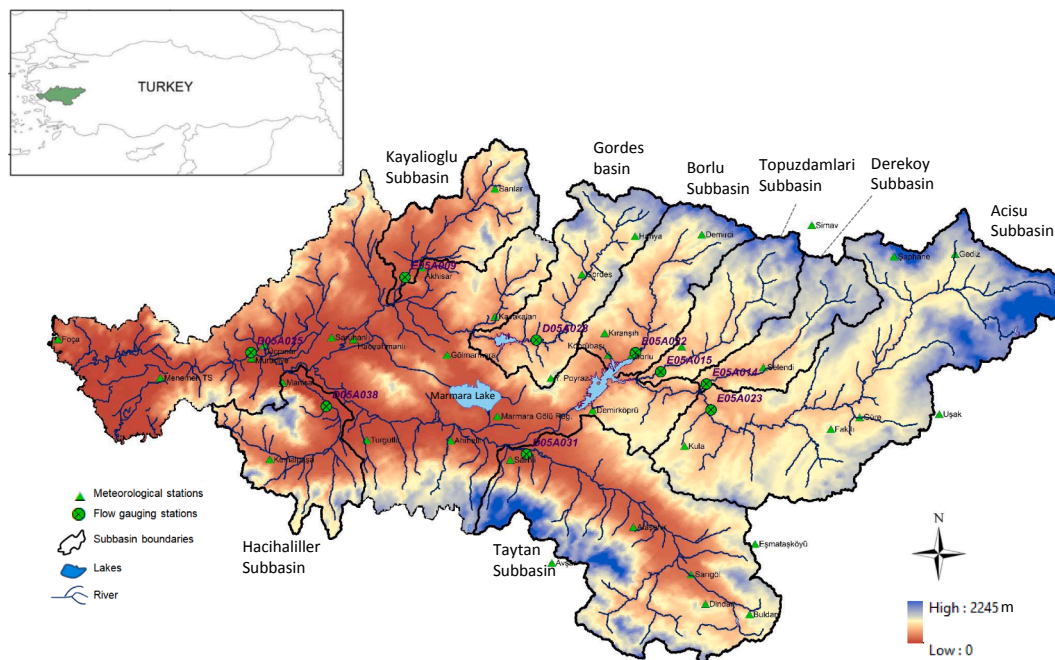


Fig. 1. Case study area, the locations of the stations, and the corresponding subbasins.

since it is a parsimonious model and performs well for different climatic conditions (e.g., Tekleab et al., 2011; Zuo et al., 2014; Li et al., 2016; Okkan and Kirdemir, 2020). The model is also practical in that it only needs monthly total precipitation (P) and potential evapotranspiration (PET) as input variables to simulate monthly runoff. One of the reasons for choosing the model is that it extends the Budyko (1958) hypothesis (or the *limits* approach) to promise a consistent framework for various hydrological processes. It is emphasized in the model that the actual evapotranspiration (E_{act}) is always less than the PET . Also, deep percolation (also called groundwater recharge) can occur even when the soil is not saturated (Li et al., 2016). A schematic diagram of the model, including computation steps, is shown in Fig. 3. Accordingly, it is assumed in the *dynwbm* that precipitation $P(t)$ in time step t is partitioned into catchment rainfall retention $C(t)$ and direct runoff $Q_{direct}(t)$. After calculating $C(t)$ based on α_1 (first Budyko-typed curve parameter) and the maximum capacity of soil moisture storage (S_{max}), $Q_{direct}(t)$ is simulated (see Figs. 3-E1&E2). In the model, the water availability for basin can also be expressed as

$$wc(t) = S(t) + E_{act}(t) + Rec(t) = C(t) + S(t-1) \quad (1)$$

where $S(t)$ and $Rec(t)$ are soil moisture content and groundwater recharge, respectively.

In the model, following the calculation of evapotranspiration opportunity $eop(t)$, which is controlled by the second Budyko-typed curve parameter α_2 termed as evapotranspiration efficiency, $Rec(t)$ is simulated (see Figs. 3-E4&E5). This parameter also controls E_{act} (see Figs. 3-E6) because the occurrence of groundwater recharge is essentially linked to the evapotranspiration process (i.e., a decrease in parameter α_2 leads to higher recharge). After those computations, the soil moisture content is updated (see Figs. 3-E7). Besides, the groundwater storage is treated as a linear reservoir (see Figs. 3-E8&E9). Here, a particular amount of the groundwater storage $G(t-1)$ from the previous month contributes as baseflow $Q_{base}(t)$. Finally, the sum of Q_{direct} and Q_{base} gives the total simulated runoff (see Figs. 3-E10).

3.2. Overview of ANN

Despite several ANN architectures documented, the most prevalent type is feed-forward neural networks termed as the multilayer perceptron occasionally (Lohani et al., 2011). In the study, the considered ANN

model consists of three layers of processing artificial neurons (or nodes), which are arranged in the input layer, hidden layer, and output layer, respectively. The computational complexity of ANN depends not only on the number of predictors but also on the number of nodes in the hidden layer (n_{nh}). Herein, while the inputs are assigned as an outcome of experience or statistical analysis, a trial-and-error approach is usually preferred when finding the appropriate n_{nh} (Noori and Kalin, 2016). In this layered architecture, the operating procedure involves two phases: namely, forward computing and training. The forward computing step was explained in detail by Fistikoglu and Okkan (2011). In training, the weights and bias terms are calibrated by optimization algorithms, which can also preferably be used to calibrate CRR models. In the study, the automatic optimization algorithm controlling the ANN training was also utilized in the CRR model calibrations (see Section 3.6). In both sole ANN models and hybrid models containing ANN, all variables were scaled to fall in the range [0,1]. Accordingly, the log-sigmoid transfer function gave appropriate results in all layers. After training, the scaled network outputs were converted back into the runoff simulations with the same unit as the original targets.

3.3. Overview of SVR

SVR is another type of ML technique that is widely applied to rainfall-runoff modeling and streamflow simulation (e.g., Lin et al., 2006; Behzad et al., 2009; Maity et al., 2010; Okkan and Serbes, 2012; Tongal and Booi, 2018). SVR includes structural risk minimization instead of empirical risk minimization used in many methods such as ANN. In our study, the least-squares version of SVR is preferred because it possesses a simplified algorithm and requires less effort in model training than the original SVR. The main premise of SVR is based on the usage of a kernel function that nonlinearly maps the input data into higher dimensional feature space (Sachindra et al., 2013). Although various kernel functions are used within the scope of SVR, it has been observed that the radial basis kernel function outperforms other types (Tongal and Booi, 2018). Besides, the data scaling approach in SVR is the same as in ANN. To get a qualified simulation from SVR, two governing parameters, which are the regularization parameter (i.e., C_p), and the width parameter of the radial basis kernel function (i.e., σ), should be adjusted. The other details of this ML technique can be found elsewhere (Anandhi et al., 2008; Okkan and Serbes, 2012; Sachindra et al., 2013; Okkan and Inan, 2015).

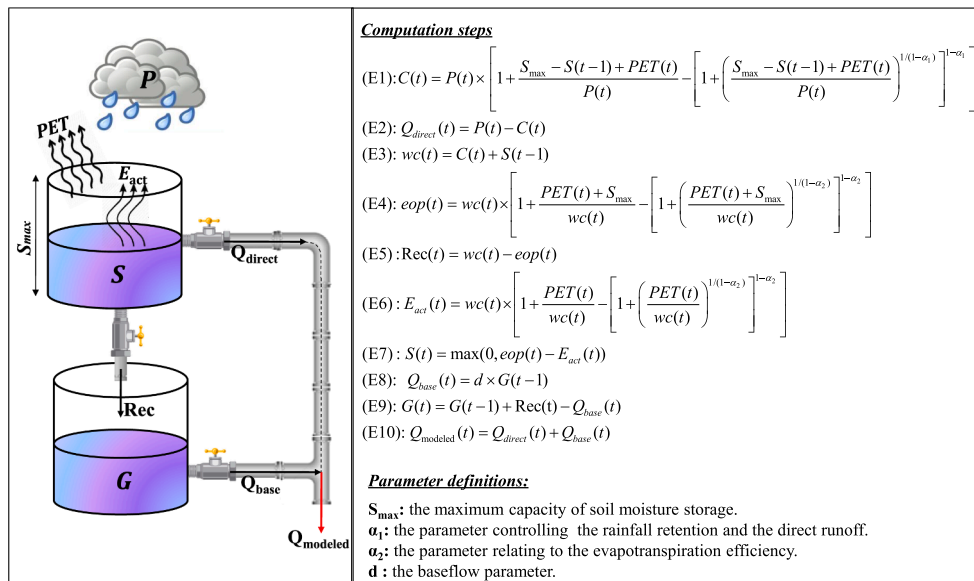


Fig. 3. Schematic diagram of the *dynwbm*, the related calculation steps and the parameter definitions; the unit of the defined variables for fluxes are millimeter month⁻¹ and millimeter for the storage elements (S and G).

In the study, sole ANN and SVR models were first trained with two inputs (i.e., $P(t)$ and $PET(t)$). Then, antecedent precipitation data [$P(t-1)$] were also included in the input set to represent the initial catchment wetness. These two-and three-input ML models were referred to as ANN1/SVR1 and ANN2/SVR2, respectively.

3.4. Nested hybrid rainfall-runoff models

The nested hybrid models were built by integrating the approaches outlined in Section 3.1, Section 3.2, and Section 3.3. As depicted in Fig. 4a, the hybrid models are consecutively composed of conceptual and ML parts. In the conceptual part that requires P and PET inputs and initial soil moisture content (S_0), the expressions denoted in Fig. 3 are executed with the conceptual parameters. Then, the various simulated outputs are transmitted to the ML part. In the first strategy, two conceptual outputs (Q_{direct} and Rec) that provide high correlations with observed runoff series are evaluated separately with the ANN and SVR techniques (relevant correlation coefficient statistics that form the basis for selecting those outputs are presented in Section 4.2). Herein, the ML

part eliminates the baseflow computation of the conceptual model, where the groundwater storage is governed with parameter d , and converts the mentioned conceptual outputs to the simulated runoff at the drainage basin outlet. The reason why the baseflow component of the conceptual model was disabled within the nested scheme was explained in Section 4.1. Depending on which ML type (ANN or SVR) is included, these nested hybrid models are referred to as dynNN1 and dynSVR1, respectively. In the second nested hybridization, a similar process is executed with a sufficient number of principal components derived from four outputs. In the principal component analysis (PCA), while the first principal component (PC) provides the most meaningful relationship in the variables directed according to the most considerable variance, the second PC, which is orthogonal to the first one, accounts for most of the remained variance (Hu et al., 2007). For the covariance matrix-based extraction procedure about PCs, the corresponding coefficients, and eigenvectors, the readers can review Rencher (1998). The nested hybrid models incorporating PCA as a data pre-processing tool are referred to as dynNN2 and dynSVR2.

As can be seen from Fig. 4a, both three parameters managing the

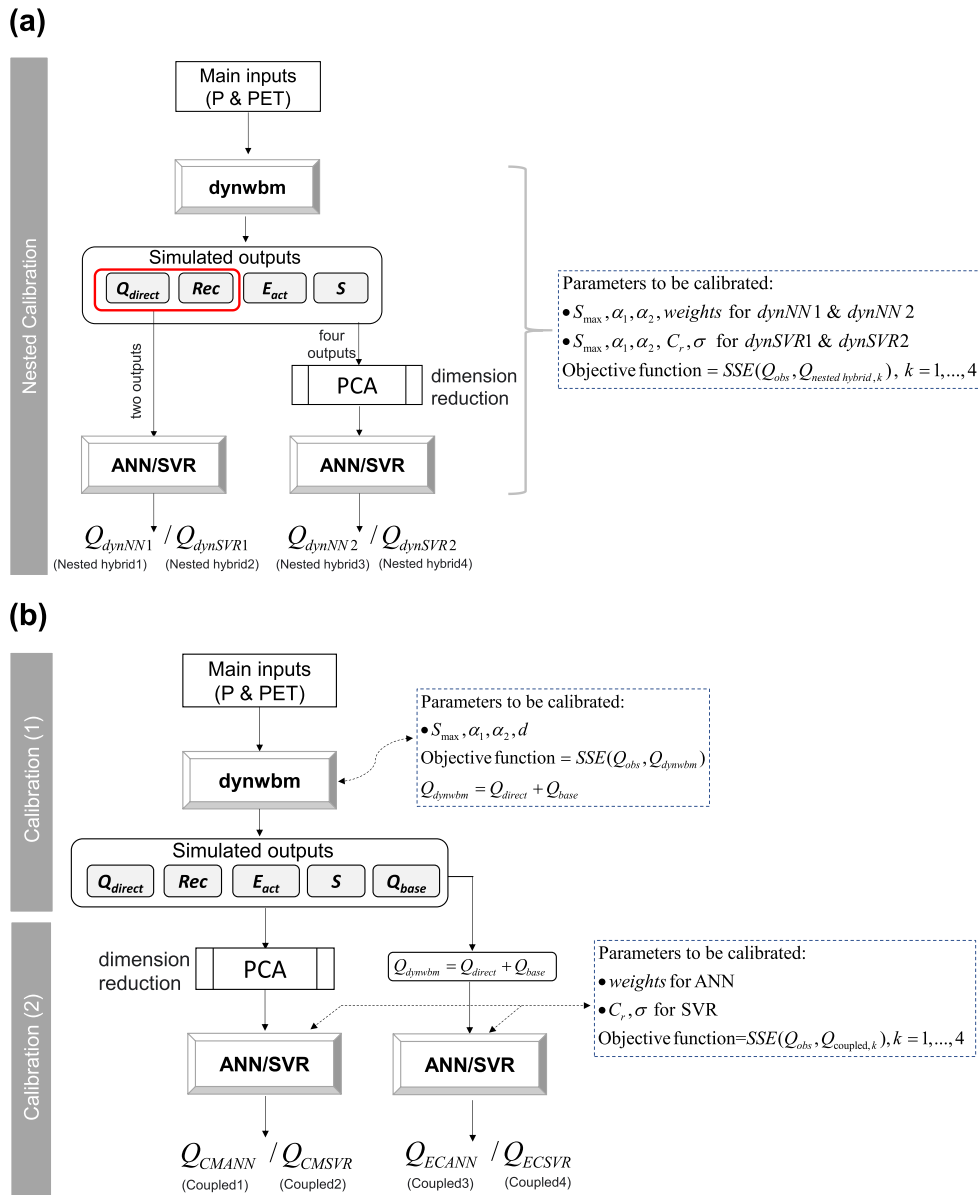


Fig. 4. (a) Nested hybrid modeling strategies operated through one calibration, and (b) parallel coupling strategies consisting of two consecutive calibration processes (objective function was operated with scaled data for all hybrid models).

conceptual part (i.e., S_{max} , α_1 , and α_2) and the weights providing inter-layer connection in the ANN part need to be tuned for the dynNN1 and dynNN2 models. In other words, the dynNN1 and dynNN2 models whose total number of free parameters is equal to $4 \times nnh + 4$ should be trained. Similarly, the same conceptual parameters, along with two parameters that govern the SVR, need to be adjusted under the dynSVR1 and dynSVR2 models. Thus, there are only five parameters to be calibrated in nested hybrid models with SVR. The training algorithm used at this stage is described in Section 3.6.

3.5. Parallel coupling strategies used

The coupling of the conceptual model outputs with the ML techniques (ANN and SVR) is also tested against four nested hybrid modeling strategies operated with one calibration. In contrast to what is practiced in the nested models, the baseflow component is included in the coupled models with the idea that PCA will provide the required dimension reduction here. The PCA, which is also assessed within the nested modeling scope, will help avoid collinearity among the five outputs (Q_{direct} , Rec , E_{act} , S , and Q_{base}) derived from the *dynwbm*. As seen in Fig. 4b, after the *dynwbm* calibration, the dimension of those outputs is reduced with PCA, and a new training process is separately executed with ANN and SVR. Herein, the number of variables directed to the input layer of both the ANN and SVR was set as two for coupling models. The parallel hybrid schemes used are termed as CMANN and CMSVR, respectively, relying on ANN or SVR inclusion.

Furthermore, additional implementations were made to examine whether the error-correction (EC) approaches, which were previously experienced by Shamseldin and O'Connor (2001), give similar results to the nested hybrid models or not. ANN and SVR are individually examined as simple output updating techniques for simulated total runoff series ($Q_{direct} + Q_{base}$) generated from *dynwbm*. These EC-based coupling models actuated with single input series are referred to as ECANN and ECSVR, respectively (see Fig. 4b).

3.6. Parameters of models

Since the probable multiple local minima existence in the models can make searching the global parameter solution complicated, the qualification of the automatic search algorithm used at the calibration stage is crucial. Gradient-based local search algorithms are insistently chosen within this concept. But, recently, their several drawbacks (e.g., getting trapped in local minima and being overly sensitive to the initial solution) have led hydrologists to metaheuristics algorithms in the automatic calibration of CRR models (e.g., Arsenault et al., 2014) and the training of ML models (e.g., Chen et al., 2015). Besides, some modifications that bring in faster convergence ability to metaheuristics have also existed. Regarding this issue, Okkan and Kirdemir (2020) have proposed an optimization algorithm, which is made up of the hybridization of particle swarm optimization and Levenberg-Marquardt algorithms. In point of various indices in their work, it has been proven that the hybrid algorithm referred to as HPSO performs more successfully in CRR model calibration compared to familiar metaheuristics. The HPSO was also evaluated in the training of all models compared. Further details about the algorithm can be accessed from Okkan and Kirdemir (2020).

In the study, the data compiled for streamflow gauging stations were divided into two equal parts for calibration (the water period of 1981–1995) and validation (the water period of 1996–2010). Then, the variables controlling the HPSO algorithm that automatically calibrate the models were assigned. The cognitive and social parameters within the algorithm were both set to 2.0. The velocity updating rule of the algorithm was controlled through the chaotic random inertia weight. Setting the population size in HPSO to 50 was sufficient for all experiments, while the number of required iterations varied depending on the case and model type. The conventional sum-of-squared errors (SSE)

function was used as the objective (fitness) function within the HPSO. The bounds in which the conceptual parameters seek optimal solutions were also determined after various trials. Accordingly, the lower–upper bound of parameter S_{max} is 50–1000 mm, while those of α_1 and α_2 are 0.1–0.90. In addition, the bound of 0.01–1.0 is used for the parameter d . For each HPSO run, not only random parameter initializations but also the iterative solutions were kept between those closed ranges.

As is known, using warm-up periods is recommended prior to calibration for conceptual models to reach an optimal initial soil moisture state (Kim et al., 2018). In some daily hydrological model applications, initial soil moisture could also be defined as a free parameter to be calibrated (e.g., Mostafaie et al., 2018). Since the models employed in the study are monthly time-scaled ones, they are not expected to be overly sensitive to the initial soil moisture store (S_0) values. In addition, the first month of the water year is October, and the rainfall regime is relatively weak for the previous month. Thus, without setting apart a warm-up period in the calibration period, S_0 values at $t = 0$ were set to zero for all cases. Our experiments demonstrated that root mean square error (RMSE) statistics did not alter significantly between $S_0 = 0$ and $S_0 = 0.2 \times S_{max}$ (not shown).

Besides, the weights and biases, which are free parameters of ANN and hybrid models with ANN, were initialized within the range $[-1, 1]$. But they were not forced to stay within any range during iterations. Any model involving ANN with a range of 2–20 hidden nodes was sensitively trained by HPSO for the calibration data set, and the suitable combination of parameters also provided the best performance in the validation period (i.e., lowest SSE or RMSE) was extracted. The selection was made from a pool of 95 simulations (five repetitions were done for each hidden node variation). In addition, C_r and σ parameters of sole SVR and hybrid models with SVR were randomly initialized within the range $[0.1, 10]$. However, no constraint was applied to SVR parameters during HPSO iterations. The optimal Lagrangian multipliers of any model with SVR were kept that minimize the objective function calculated for the calibration data set. Afterward, like the practice in ANN-based models, the best combination of parameter sets that resulted the lowest RMSE was stored.

3.7. Criteria used for evaluating the performances of models

In addition to questioning SSE or RMSE during training and validation periods, Nash–Sutcliffe efficiency (NS) was used. The NS (Nash and Sutcliffe, 1970) is commonly preferred in assessing performances of hydrological models, and as emphasized by Moriasi et al. (2007), it is a consistent index for reflecting the overall fit of a hydrograph. The models were graded in terms of their NS performance as follows (Moriasi et al., 2007; Noori et al., 2020):

Very good: $0.75 < NS \leq 1$
 Good: $0.65 < NS \leq 0.75$
 Satisfactory: $0.50 < NS \leq 0.65$
 Unsatisfactory: $NS \leq 0.50$

In the study, to what extent the implemented models differ from the *dynwbm* was also investigated by NS-based r^2 statistics recommended by Nash and Sutcliffe (1970). This index can be expressed by the following equation:

$$r^2 = \frac{NS_2 - NS_1}{1 - NS_1} \quad (2)$$

where NS_1 and NS_2 are the NS coefficients obtained from *dynwbm* and any implemented model, respectively.

According to Senbeta et al. (1999), r^2 greater than 10% indicates that the proposed model showed a significant improvement compared to the reference model. Although we are aware that there are other performance measures, it is intended to provide a straightforward comparison

due to the large number of models to be compared in the study (also, since the rankings of the various options turned out to be almost the same, they were not shown).

4. Results

In this study, the computer codes were developed in MATLAB® environment. All the models were implemented for nine streamflow gauging stations. To explicate whether the nested hybridizations influenced simulation performances, sole models and different coupled models were also then developed. But first, a sensitivity quantification applied to *dynwbm* that suggests using groundwater recharge output instead of the baseflow component within nested hybrid models was presented in Section 4.1. Then, how to set up nested hybrid models is explained in Section 4.2. Finally, the comparison of all candidate models is introduced in Section 4.3.

4.1. Parameter sensitivity analysis of *dynwbm*

Following the calibration and validation of the *dynwbm*, a variance decomposition sensitivity analysis-based framework termed ANOVA was operated to dynamically measure the individual and interactive impacts of *dynwbm* parameters on the runoff simulation performance. Qi et al. (2016) previously suggested a three-way ANOVA framework to quantify the dynamic sensitivity of an optimization algorithm's control

parameters and investigated the parameter influences over a hydrological model calibration problem. In the study, the global sensitivity analysis exerted to four-parameter *dynwbm* can be referred to as four-way ANOVA. Before adapting ANOVA to parameter sensitivity analysis, the parameter combinations denoted in Fig. 5a were formed, and the *dynwbm* was run $17^4 = 83521$ times in total. The SSE variations collected from those runs were then partitioned into four main effects corresponding to four parameters of *dynwbm*. In addition, there are six first-order interaction terms (i.e., $S_{max}*\alpha_1$, $S_{max}*\alpha_2$, $S_{max}*d$, $\alpha_1*\alpha_2$, α_1*d and α_2*d), four second-order interaction terms (i.e., $S_{max}*\alpha_1*\alpha_2$, $S_{max}*\alpha_1*d$, $S_{max}*\alpha_2*d$, and $\alpha_1*\alpha_2*d$) and one third-order interaction term (i.e., $S_{max}*\alpha_1*\alpha_2*d$), describing the situation where effects are nonlinear. After decomposing the partial variances through four-way ANOVA, the contributing variance fractions related to main sources and interaction terms were obtained for each case. Finally, the average values of these variance fractions across the basin are summarized in Fig. 5b. It is clear from Fig. 5b that the sensitivity of α_1 and α_2 parameters on the *dynwbm* performance is more dominant, while the parameter d , which linearly controls the baseflow, lacks sensitivity. The contributions of first-order interaction terms based on the parameter d in the total interaction of 7% were also extracted in Fig. 5c. Accordingly, the contributions of these interaction terms referred to as $S_{max}*d$, α_1*d , and α_2*d within the total interaction are again relatively ineffective (second and third-order interaction contributions are rather weak and are not shown in the figure).

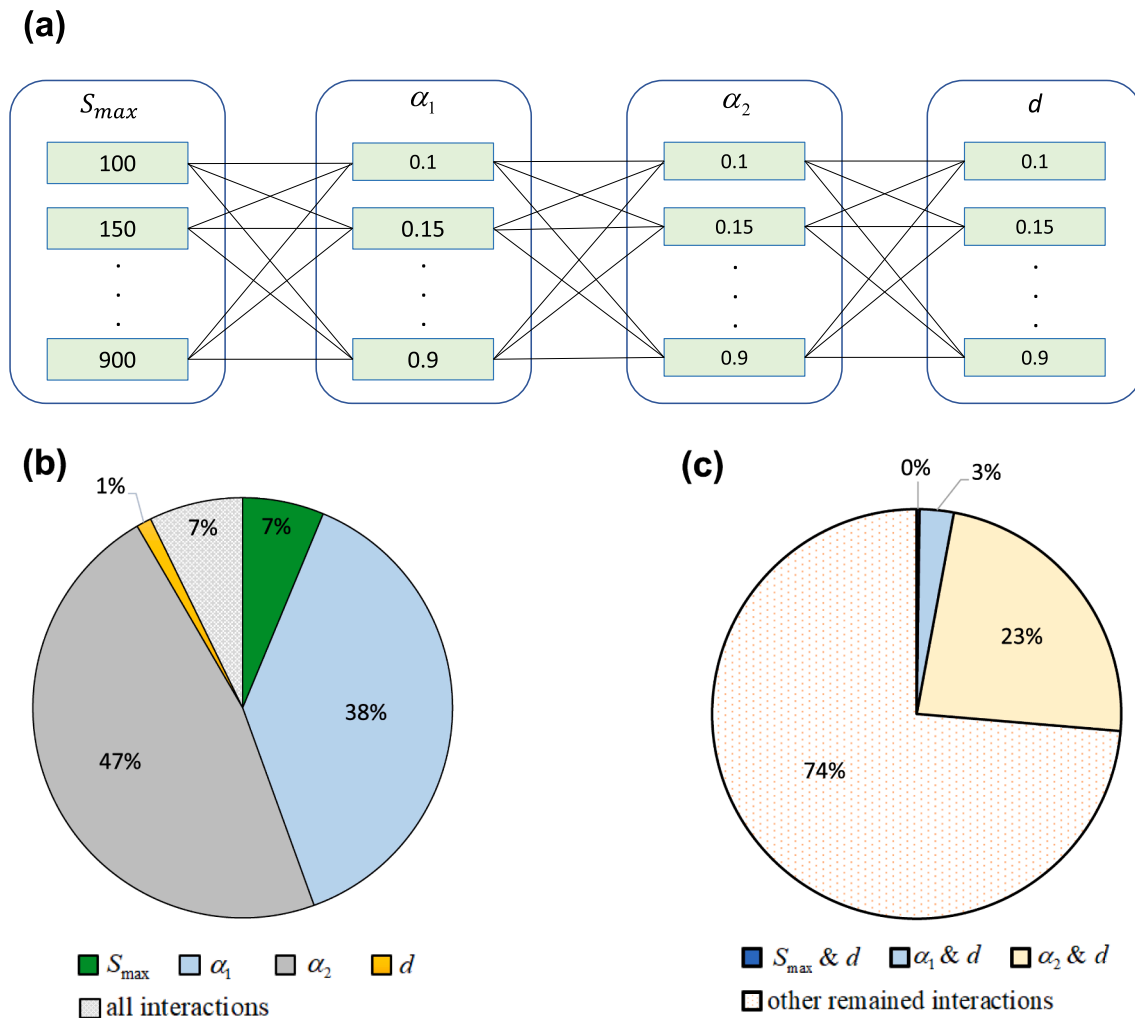


Fig. 5. (a) Chain diagram representation for the combinations of *dynwbm* parameters (S_{max} in millimeter), (b) variance fractions for main sources and total interaction, and (c) variance fractions of first-order interactions regarding parameter d and the other remained fractions for eight interaction terms.

4.2. Setting up nested hybrid models

The findings in Section 4.1 first prompted us to discuss whether some modifications or improvements could be made for the baseflow function. To this end, first, considering the nonlinear aquifer storage-discharge relationship recommended by Gan and Luo (2013), the linear baseflow form was turned into a power function with two parameters. However, this modification did not yield adequate results during the validation period (not shown). Therefore, as mentioned in Section 3.4, whether machine learning hybridization like ANN/SVR, which can substitute for the groundwater storage element of *dynwbm*, gave more reasonable simulations were questioned. Some statistical analyses were applied to the simulated outputs of the calibrated *dynwbm* to have an idea about how to hybridize the model. For instance, the correlation coefficients (R) between certain internal variables of the conceptual model and the observed runoff (Q_{obs}) were derived in Fig. 6. According to the boxplot graph pertaining to the R statistics obtained for all stations, Rec and Q_{direct} offer the highest correlation with Q_{obs} , respectively. Even though soil moisture content S is another variable that provides a high correlation with Q_{obs} , it is unnecessary to direct it to the ML part within the hybrid scheme since it is fairly correlated with Rec . In the light of these indications, in the first nested approach, the ML part only processes the Rec and Q_{direct} outputs of the conceptual part that are governed with three parameters. In this scheme, ANN and SVR were used as ML techniques, respectively, and these nested hybrid models were attributed as *dynNN1* and *dynSVR1*, respectively.

In the other nested scheme, not only the Q_{direct} and Rec outputs of the conceptual model but also S and E_{act} outputs were operated within the ML part. However, since the experimental analysis denoted in Fig. 6 implies that strong inter-correlations exist between some model outputs, multi-collinearity originating from these inter-correlations may lead to poor generalization performance of the ML part. In addition, sending many signals to the input layer of the ML part also complicates the calibration process. In this perspective, as addressed in Section 3.4, there is a requirement for a data pre-processing technique that can convert the correlated conceptual outputs into uncorrelated ones and reduce the dimensionality of those while not losing important information. In the

study, before directing those four outputs to the ML part, it was found appropriate to expose them to PCA. In those models referred to as *dynNN2* and *dynSVR2*, the first two PCs explained a large proportion of the variance for each case (not shown). As a result, the number of variables directed to the input layer of the ML part was set as two.

4.3. Intercomparison of models

In the study, the calibration period findings (not shared) indicate that the increase of the number of parameters in hybrid models has also reflected on the simulation accuracy. Therefore, it is thought that it will be more reasonable to scrutinize the relative performances of all candidate models with regard to the validation period simulations only. The simulation accuracy results for all the models calibrated when applied to the validation data from 1996 to 2010 (months October to September) based on all flows (180 data points) are presented in Table 2. According to NS indices stated in Table 2b, each model performance, which was rated in terms of the performance grading recommended by Moriasi et al. (2007), was summarized in Fig. 7a as well. It is clear from this figure that the *dynwbm* provides *very good* simulation on four stations, while the NS values of the remaining five stations only fall into the *good* category. However, when the NS performances of two-input ANN and SVR models were examined, it was observed that no stations with *good* and *very good* ratings existed, and five stations were rated as *satisfactory*, whereas the rest were graded as *unsatisfactory*. On the other hand, ANN and SVR models with three inputs gave better results than those with two-input ML variants. In the ANN2, which is slightly better than SVR2, it was determined that eight stations were *very good* or *good*-rated, while one station was *satisfactory*-rated. When the coupled model variants were investigated, it was seen that performances compiled from all stations were rated as *good* or *very good*. Two stations with *good*-rated performances in *dynwbm* switched to the *very good* category with error-correction based models (i.e., ECANN and ECSVR). Besides, *dynwbm*-ANN coupled model (CMANN) and *dynwbm*-SVR coupled model (CMSVR), which give slightly preferable performances than those error-correction types, were classified as *very good* except for a few stations. Moreover, it is immediately clear from the same figure that each of the

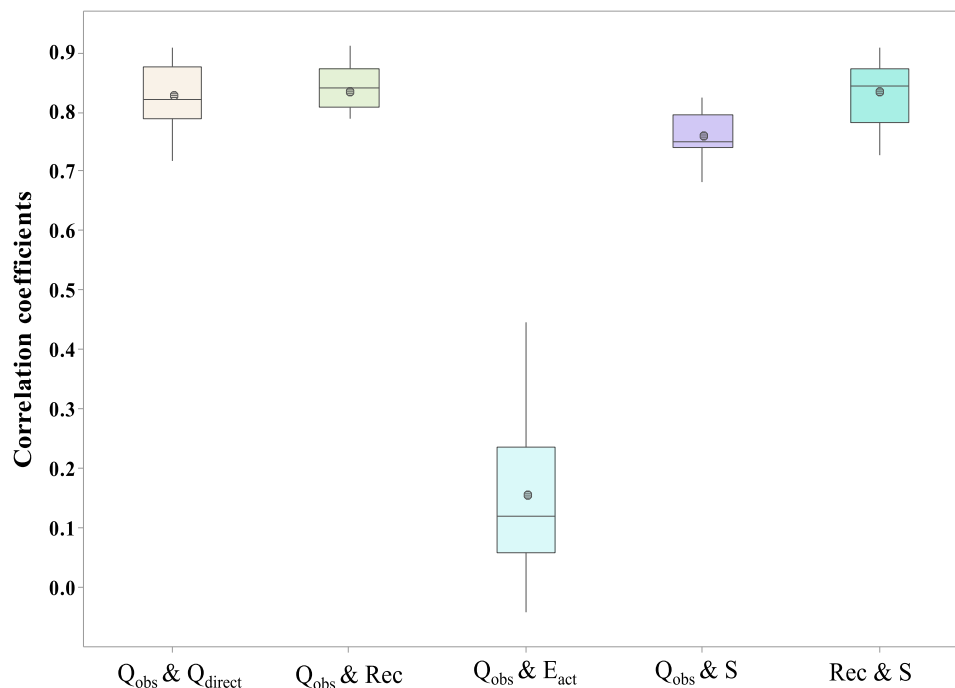


Fig. 6. Boxplot graph generated with the R statistics obtained for all stations. (the horizontal lines in the middle of the boxes and the circles represent the median and mean, respectively. The edges of the box are the 25th and 75th percentiles, and the whiskers extending to the extreme data points are also shown).

Table 2

Performance comparison of all models considered for the validation period on each streamflow gauging station based upon all flows: (a) RMSE statistics (in millimeters), (b) NS indices.

(a)													
Stations	<i>dynwbm</i>	<i>ANN1</i>	<i>SVR1</i>	<i>ANN2</i>	<i>SVR2</i>	<i>ECANN</i>	<i>ECSVR</i>	<i>CMANN</i>	<i>CMSVR</i>	<i>dynNN1</i>	<i>dynSVR1</i>	<i>dynNN2</i>	<i>dynSVR2</i>
Muradiye	4.64	6.20	6.28	4.59	4.53	4.31	4.04	3.27	3.37	3.07	2.70	2.80	2.64
Hacihidir	8.34	14.42	14.89	9.34	9.69	8.04	8.05	9.21	8.53	<u>7.03</u>	7.35	7.14	7.21
Taytan	3.40	5.38	5.37	3.95	3.95	3.42	3.35	3.37	3.14	3.03	2.91	3.07	2.94
Hacihaliller	8.19	8.95	9.24	6.26	6.50	7.61	7.62	6.90	7.08	5.34	4.95	5.06	4.63
Kayalioglu	7.43	8.84	9.17	5.89	6.99	6.63	6.43	6.39	6.53	4.93	4.82	4.74	4.59
Derekoy	5.82	7.63	7.46	6.20	5.73	5.79	5.44	5.21	5.26	4.18	3.56	4.88	3.93
Topuzdamlari	7.71	12.49	12.37	10.74	10.43	7.28	6.61	7.19	6.59	7.04	5.90	7.41	6.09
Borlu	6.24	10.56	10.68	6.99	7.54	6.23	5.91	6.72	6.97	5.11	4.69	4.80	4.37
Acisu	4.83	6.31	6.31	4.71	4.75	4.07	3.98	3.68	3.56	3.50	3.35	3.42	3.31
(b)													
Stations	<i>dynwbm</i>	<i>ANN1</i>	<i>SVR1</i>	<i>ANN2</i>	<i>SVR2</i>	<i>ECANN</i>	<i>ECSVR</i>	<i>CMANN</i>	<i>CMSVR</i>	<i>dynNN1</i>	<i>dynSVR1</i>	<i>dynNN2</i>	<i>dynSVR2</i>
Muradiye	0.735	0.526	0.514	0.740	0.748	0.771	0.799	0.868	0.860	0.884	0.910	0.903	0.914
Hacihidir	0.837	0.513	0.481	0.796	0.780	0.849	0.848	0.801	0.830	0.884	0.874	0.881	0.878
Taytan	0.798	0.493	0.494	0.726	0.727	0.794	0.803	0.801	0.827	0.839	0.852	0.835	0.849
Hacihaliller	0.666	0.601	0.575	0.805	0.789	0.712	0.710	0.762	0.750	0.858	0.878	0.872	0.893
Kayalioglu	0.654	0.509	0.472	0.782	0.693	0.724	0.740	0.743	0.732	0.847	0.858	0.859	0.868
Derekoy	0.708	0.498	0.521	0.669	0.717	0.711	0.745	0.766	0.762	0.849	0.891	0.795	0.867
Topuzdamlari	0.808	0.497	0.506	0.628	0.649	0.829	0.859	0.833	0.860	0.840	0.888	0.823	0.880
Borlu	0.855	0.586	0.576	0.819	0.789	0.856	0.870	0.832	0.820	0.903	0.918	0.914	0.929
Acisu	0.688	0.467	0.468	0.703	0.698	0.778	0.788	0.819	0.830	0.836	0.850	0.844	0.854

While the top three models in terms of performance indices assessed are shown with bold numbers for the related station, the best one among them is indicated by underlined numbers.

nested hybrid models has performed in the *very good* category at all stations.

The RMSE, a measure of overall error between the observed and simulated runoff, was also computed for each case, as seen in Table 2a. Despite station-based variations pertaining to this measure, if we interpreted those with averages across the basin, nested hybrid models achieved reductions in RMSE in the range of 24% and 30% compared with *dynwbm*. When the same assessment is made with coupled models except for ECANN, they gave roughly 10% lower RMSE than that calculated for the conceptual model. Unlike observed in coupled and nested hybrid models, a reduction in RMSE was not observed in ML models, excluding some notable reductions achieved with ANN2 and SVR2.

After processing the NS indices in Table 2b with Eq.2, the significances of r^2 values can be seen visually in Fig. 7b. Based on NS performances obtained with *dynwbm*, it is seen that ANN2/SVR2 did not provide a substantive development and did not have the potential to haul the relevant indices to a better category. Thus, both ANN1 and SVR1 were not even included in this comparison. It is also obvious from this figure that nested hybrid models tend to progress remarkably in terms of r^2 at all stations, excluding only the insignificant r^2 value below 10% obtained by the dynNN2 at Topuzdamlari station. Therein, values given on the figure with a green color scale indicate that the hybridization yielded a noteworthy improvement. All or some of the coupled models displayed either deterioration or inconsiderable performance increases at some stations such as Hacihidir, Taytan, Derekoy, and Borlu. It can be affirmed that CMSVR is a bit more prominent among other parallel hybridization variants for many stations. However, negative values of r^2 observed for Hacihidir and Borlu stations signify that this coupling lead to undesirable responses in the validation data. In this regard, it may seem favorable that ECSVR does not offer a negative r^2 , at least in any station. It even suggested a significant improvement in Derekoy and Borlu stations compared to its counterpart ECANN. Among all hybridization exercises, it has been detected that the models that provide the most remarkable improvements for entire stations are dynSVR2 and dynSVR1.

The above results were reinforced by boxplot demonstrations to visually summarize how the models performed in terms of the general NS distributions over the basin. It can be seen from Fig. 8a, prepared with NS performances based on all flows compiled for all stations, that

interquartile ranges (IQRs) for the performances obtained from all nested hybrid models and two coupled models (i.e., CMANN and CMSVR) are entirely located above the 0.75 (lower limit of *very good* model). On the other hand, the plots resulting from the dynSVR1 and dynSVR2 models are almost identical and indicate less interstation NS variability over the basin on account of their narrower IQRs. Compared with the other nested hybrid models and all single or coupled models, operating these models resulted in superior median and quartile accuracy over the basin. Although both the median and upper quartile accuracy associated with dynNN2 is superior to those of dynNN1 over the basin, dynNN1 is somewhat more credible, particularly at the lower quartile of the distribution. On the other hand, dynNN1 is slightly positively skewed, unlike what is seen in other nested hybrid models. It is also clear from Fig. 8a that *dynwbm* and coupled models including ECANN, ECSVR, and CMSVR displayed approximately 3–4 times broader IQRs than those of dynSVR2 and dynSVR1. Since it is known that these IQRs consist of 50% of the corresponding performance distributions, it can be asserted that these variants may not behave uniformly across the basin and therefore could not reveal performance consistency as warranted in these two nested hybrid models.

In the study, accurate high flow simulation is particularly important at some stations (Derekoy, Topuzdamlari, Borlu, Acisu, and Taytan), which act on agriculture-operated reservoirs of the basin, and at the Hacihidir station linked with freshwater management for Gordes reservoir. Therefore, the indices given in Table 2 were interpreted again based on the runoff simulations greater than the 75th percentile observed runoff for each station and presented in Table 3. This evaluation was made for the same 45 data points at all stations. Table 3 reveals that when interpreted for high flows, the nested hybrid models again tend to provide the most accurate results. In fact, the first three model rankings in Table 2 and Table 3 are rather identical. In the majority of stations, the dynSVR1 and dynSVR2 produced results that could fall into the *good* or *very good* category even at these extreme points, which represent one-quarter of the total validation data. Besides, at these extreme points, it was observed that the solo *dynwbm*, ANN2, and SVR2 models failed to generate results in the *good* or *very good* category, with a few exceptions. Especially, the extrapolation capability of *dynwbm* does not seem good, and combining this model with CMANN did not contribute to the grading in high runoff simulation. On the other side, for high flows observed at Muradiye, Hacihidir, and Topuzdamlari

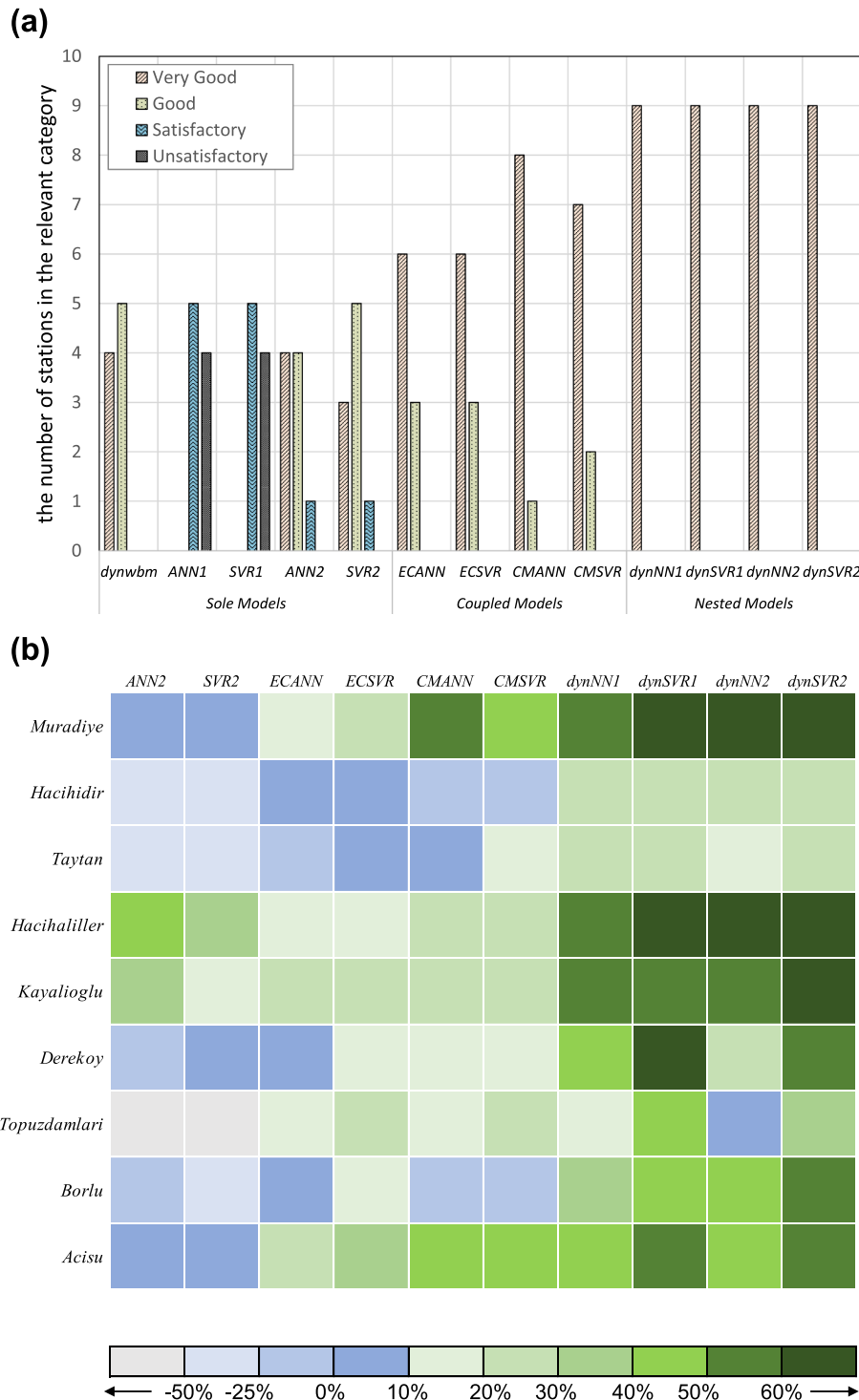


Fig. 7. (a) The number of streamflow gauging stations with the relevant performance ratings according to the models and (b) r^2 statistics displaying NS-based improvements or deteriorations.

stations, the other coupling methods (ECSVR, CMSVR, and ECANN) made it possible to step up above the category offered by *dynwbm*. Nevertheless, all coupled models displayed no sign of performance progression for the upper extreme data associated with Hacihaliller, Kayalioglu, and Derekoy stations as opposed to those derived from *dynSVR1* and *dynSVR2*. Moreover, performance ratings of all models at high flows are reflected in Fig. 8b. Compared to boxplots shown in Fig. 8a, it is obvious that there are some distributional shifts over the basin in terms of simulation performances for high flows. For example,

therein, coupled models suggested 0.25–0.30 less median compared to their all flows-based performances across the basin. The mentioned median decrease is in the order of 0.17 for nested hybrid models, and the relative widening in IQRs offered through these models are less than detected in other models.

In the final stage of this section, a hierarchical clustering process was performed for the models according to the NS performances to clarify which models were more similar to each other or differed from each other. The dendrogram, shown in Fig. 9, was obtained from the complete

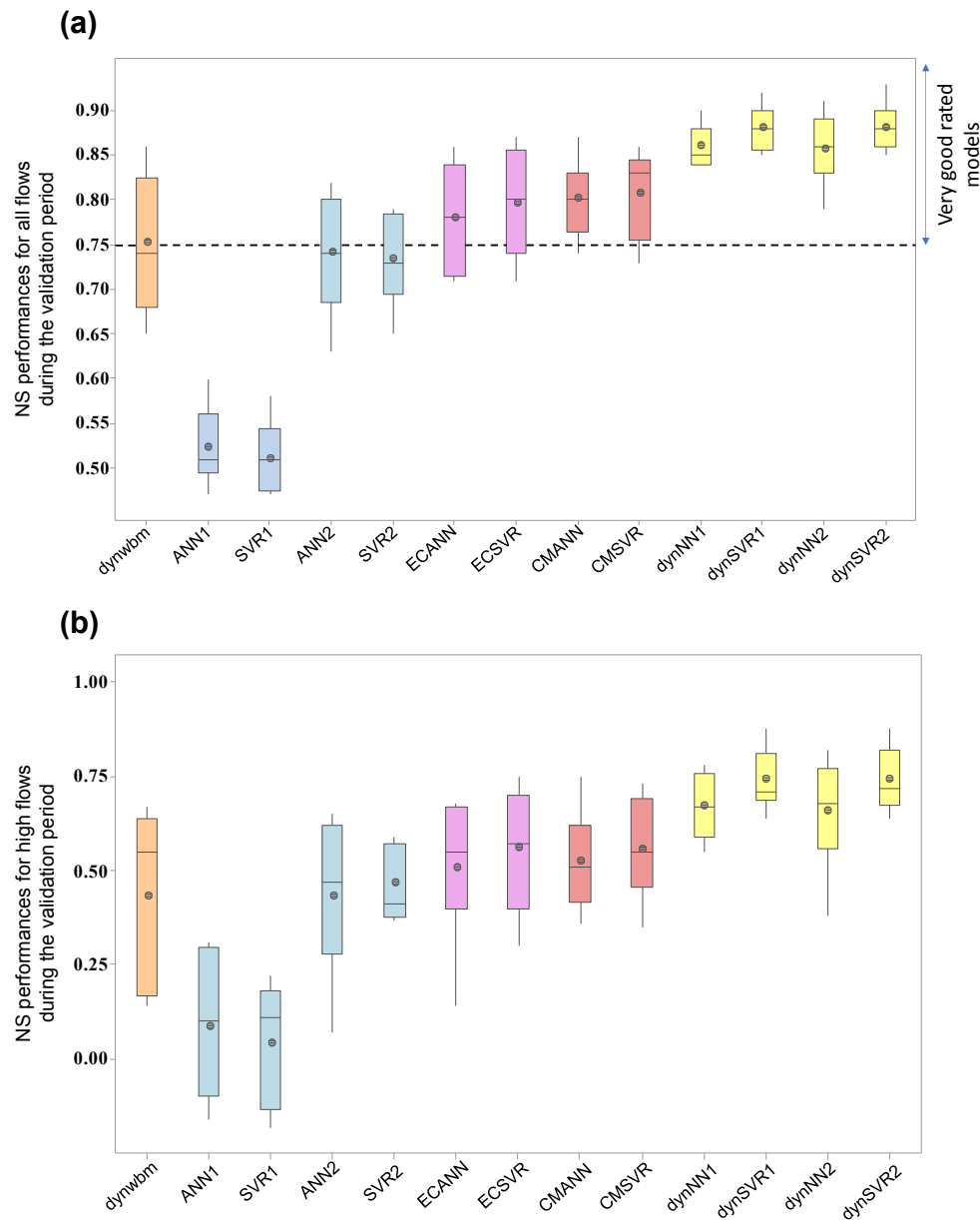


Fig. 8. Comparison of the different models for the validation period across the basin in terms of (a) all flows-based NS indices and (b) high flows-based NS indices.

linkage algorithm-based clustering procedure, and it was iteratively formed by means of three clusters, whereby each branch involves a group of similar models. Accordingly, the first cluster containing unqualified ML models with two inputs was substantially different from all the other branches of the dendrogram. The second cluster was composed of seven models, including ML models with three inputs, the sole conceptual model, and coupled models. Of course, coupled models have shown some improvements on *dynwbm*, and in this sense, *dynwbm* can be labeled as a separate branch in the third cluster. However, it can also be seen from Fig. 8a that it exhibits a similar performance distribution with models such as ECANN and ECSVR to a certain extent. Besides, the third cluster consists of all nested hybrid models. In this cluster, two subgroups turned up depending on the corresponding ML technique within the nested scheme.

In this section, it was deeply examined whether the simulations obtained from coupled hybrid models of which similar versions were assessed in the latest literature match those generated from serial nested ones that we proposed. As a result, the obtained outcomes confirmed

that neither the parallel hybridization trials (i.e., CMANN, CMSVR, ECANN, and ECSVR) nor the other sole models could take the place of nested hybrid models.

5. Conclusions

Compared with the ML models, a known profit of the CRR models is that they can simulate the physical process with a series of conceptual storages containing simple relationships and can represent wetness conditions of the basins. However, as seen in this study, systematic biases originated from the simple relations can be found in the runoff outputs derived from these models. Even though various coupling model exercises in the literature aim to correct those biases, these models never question whether optimizing both the conceptual parameters and ML parameters yields relatively better runoff simulation. In this study, ML techniques substitute for the groundwater storage part of a lumped CRR model and operate various conceptual outputs as predictor set within the conflated scheme. To the authors' knowledge, the nested hybrid

Table 3

Performance comparison of all models considered for the validation period on each streamflow gauging station based upon runoff values > the 75th percentile observed runoff: (a) RMSE statistics (in millimeters), (b) NS indices (45 data points were evaluated for each station).

(a)													
Stations	<i>dynwbm</i>	<i>ANN1</i>	<i>SVR1</i>	<i>ANN2</i>	<i>SVR2</i>	<i>ECANN</i>	<i>ECSVR</i>	<i>CMANN</i>	<i>CMSVR</i>	<i>dynNN1</i>	<i>dynSVR1</i>	<i>dynNN2</i>	<i>dynSVR2</i>
Muradiye	7.02	8.71	9.23	7.59	6.93	6.55	5.79	5.24	5.45	4.86	<u>3.61</u>	4.43	3.65
Hacihidir	15.67	25.50	27.04	16.29	16.56	14.71	14.79	17.30	15.06	<u>13.1</u>	13.85	<u>13.28</u>	13.62
Taytan	5.06	8.34	8.40	6.56	6.12	5.18	5.06	5.57	4.99	<u>4.98</u>	<u>4.67</u>	5.30	<u>4.61</u>
Hacihaliller	14.08	14.47	15.69	10.45	11.46	13.14	13.27	12.69	12.95	9.75	<u>9.22</u>	<u>9.08</u>	<u>8.54</u>
Kayalioglu	14.25	14.69	15.92	9.80	12.37	12.18	12.07	12.32	12.59	9.23	<u>8.85</u>	<u>8.86</u>	<u>8.39</u>
Derekoy	10.41	11.73	10.58	10.84	8.83	10.40	9.39	8.98	8.10	<u>7.23</u>	<u>5.61</u>	8.84	<u>6.27</u>
Topuzdamlari	13.16	21.03	20.35	18.87	17.01	12.67	<u>11.00</u>	13.40	11.83	12.84	<u>10.01</u>	13.12	<u>10.23</u>
Borlu	11.56	16.77	18.11	11.93	12.92	11.49	10.82	12.86	13.52	9.48	<u>8.70</u>	<u>8.88</u>	<u>7.85</u>
Acisu	7.96	9.20	9.42	7.43	6.73	6.61	5.87	6.14	6.12	5.85	<u>4.88</u>	5.61	<u>5.08</u>
(b)													
Stations	<i>dynwbm</i>	<i>ANN1</i>	<i>SVR1</i>	<i>ANN2</i>	<i>SVR2</i>	<i>ECANN</i>	<i>ECSVR</i>	<i>CMANN</i>	<i>CMSVR</i>	<i>dynNN1</i>	<i>dynSVR1</i>	<i>dynNN2</i>	<i>dynSVR2</i>
Muradiye	0.549	0.304	0.219	0.472	0.560	0.606	0.693	0.748	0.727	0.784	<u>0.880</u>	<u>0.820</u>	<u>0.878</u>
Hacihidir	0.628	0.014	-0.109	0.597	0.584	0.672	0.668	0.546	0.656	<u>0.740</u>	0.709	<u>0.733</u>	<u>0.719</u>
Taytan	0.573	-0.163	-0.180	0.280	0.374	0.551	0.571	0.482	0.583	<u>0.585</u>	<u>0.635</u>	0.530	<u>0.644</u>
Hacihaliller	0.325	0.286	0.161	0.628	0.552	0.412	0.400	0.451	0.429	0.676	<u>0.710</u>	<u>0.719</u>	<u>0.751</u>
Kayalioglu	0.170	0.118	-0.036	0.608	0.375	0.393	0.405	0.380	0.353	0.652	<u>0.680</u>	<u>0.680</u>	<u>0.713</u>
Derekoy	0.140	-0.091	0.113	0.069	0.382	0.142	0.300	0.361	0.480	<u>0.586</u>	<u>0.750</u>	0.380	<u>0.688</u>
Topuzdamlari	0.649	0.104	0.161	0.279	0.414	0.675	<u>0.755</u>	0.636	0.717	0.666	<u>0.797</u>	0.651	<u>0.788</u>
Borlu	0.674	0.315	0.201	0.653	0.593	0.678	0.715	0.597	0.554	0.781	<u>0.815</u>	<u>0.808</u>	<u>0.850</u>
Acisu	0.168	-0.111	-0.165	0.275	0.406	0.427	0.547	0.506	0.509	0.551	<u>0.687</u>	<u>0.587</u>	<u>0.661</u>

While the top three models in terms of performance indices assessed are shown with bold numbers for the related station, the best one among them is indicated by underlined numbers.

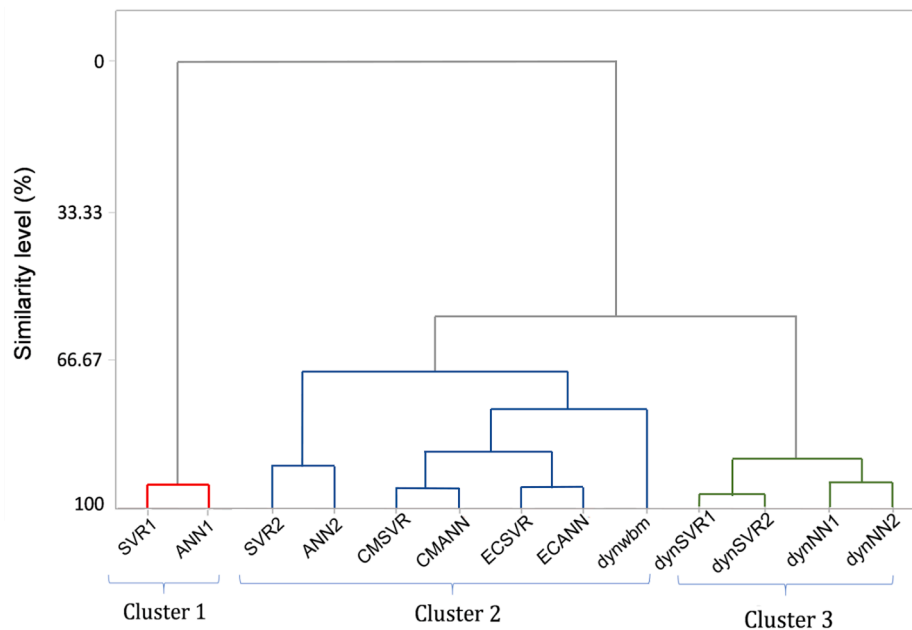


Fig. 9. Hierarchical clustering of models developed for the validation period based on NS performances.

modeling has not yet been found anywhere and stands as a true hybrid model due to its aspects mentioned above. According to some metrics assessed, all nested hybrid models and especially *dynSVR2* and *dynSVR1*, were indeed found to give the most reliable simulations for both all flows and the high flows, which are crucial for sustainable irrigation of the region. Rather than choosing between different kinds of standalone models or coupling them in a parallel approach, the nested hybridization of respective modeling approaches can serve as a useful tool for operational water resources management (e.g., streamflow forecasting and reservoir operation projections under emission scenarios). Both relative performance capabilities of the proposed model structures in runoff simulation and being exempt from the two consecutive calibrations required in coupling models may encourage other

researchers to use nested hybrid models. The pros and cons of the hybrid models proposed, and other points discussed are listed below.

- The fact that the calibrated nested hybrid models do not require any antecedent observed data [i.e., $Q(t-1)$, $Q(t-2)$] makes them sufficient to be used in the practices pertaining to the projection of future streamflow, reservoir operation, and water allocation under climate change scenarios.
- While an improvement in monthly runoff simulation was achieved with any nested hybrid model, the main conceptual structure of the reference model was widely preserved so that the soil–water fluxes, including actual evapotranspiration, recharge, and soil moisture, may serve as input data for the other exercises. For example, [Elci](#)

et al. (2010) evaluated the recharge rates obtained from a monthly conceptual rainfall-flow model as input in MODFLOW and developed a numerical groundwater flow model for a diffuse-pollution prone watershed. In another study, Trambauer et al. (2014) used the actual monthly evapotranspiration and soil moisture outputs of a hydrological model to discuss evapotranspiration deficit and root stress anomaly indices and to characterize agricultural drought in South African regions. With the proposed nested hybrid modeling strategies, auxiliary information can be provided to such applications other than streamflow or reservoir operation simulations.

- The nested hybrid models with ANN (dynNN1 and dynNN2) tend to have many parameters to be calibrated, and, eventually, it is more likely to be over-parameterized. Alternatively, this issue can be avoided by SVR, which has only two parameters to be tuned, within the hybrid scheme instead of the multilayer ANNs. Thus, nested hybrid models with SVR (dynSVR1 and dynSVR2) can be evaluated as parsimonious models with five parameters. Nevertheless, embedding both ML techniques into *dynwbm* has led to the emergence of more sensitive parameters. Therefore, the models should be run in a reasonable number of repetitions using a robust global optimization algorithm.
- In the future work of ours, the developed approaches will build a basis for the joint evaluation of different conceptual models (e.g., daily semi-distributed models) and more up-to-date ML techniques (e.g., emotional neural networks and relevance vector machines). In that possible work, we also intend to analyze the simulation uncertainty arising from the different conceptual outputs forwarded to considered ML part of the nested entity through the generalized likelihood uncertainty estimation method (see Tongal and Booi, 2017).
- It should also be noted that nested hybridization is not alleged to be an alternative for entire hydrological process modeling. In fact, it is a novel framework that admits any hydrological process inputs or outputs to enhance streamflow simulation without a necessity for double calibration as in coupling methods.

CRedit authorship contribution statement

Umut Okkan: Conceptualization, Supervision, Methodology, Software, Data curation, Formal analysis, Writing - original draft, Writing - review & editing, Visualization, Investigation. **Zeynep Beril Ersoy:** Software, Formal analysis, Writing - review & editing, Data curation, Visualization, Investigation. **Ahmet Ali Kumanlioglu:** Methodology. **Okan Fistikoglu:** Conceptualization, Supervision, Methodology, Writing - review & editing, Writing - original draft, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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